Gemini QuickDraw Clone - Production Ready Streamlit App

This document contains the final production-ready implementation of the Gemini QuickDraw Clone project. The application demonstrates multimodal GenAI integration using Gemini Vision and Streamlit.

Features Included

• Multiple rounds
• AI guessing timer
• Improved synonym matching
• Safe JSON parsing
• Production-ready architecture
• Robust Gemini integration

Architecture Flow

App Start → Generate Words → User Draws → Submit Drawing → Gemini Guess → Evaluation → Score Update → Next Round

Final app.py Code

```
# ==============================
# Gemini QuickDraw Clone
# Production-Ready Streamlit App
# ==============================

import os
import time
import json
import base64
import re
from io import BytesIO

import streamlit as st
from streamlit_drawable_canvas import st_canvas
from PIL import Image

import google.generativeai as genai

genai.configure(api_key=os.getenv("GEMINI_API_KEY"))

TEXT_MODEL = "gemini-1.5-flash"
VISION_MODEL = "gemini-1.5-flash"

TOTAL_ROUNDS = 5

SYNONYMS = {
    "car": ["automobile", "vehicle"],
    "cat": ["kitty", "kitten"],
    "dog": ["puppy"],
    "cup": ["mug"],
    "house": ["home"],
    "tree": ["plant"],
    "phone": ["mobile"],
}

def safe_json_parse(text: str):
    try:
        match = re.search(r"\{.*\}", text, re.DOTALL)
        if not match:
            return None
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            return json.loads(match.group())
        except Exception:
            return None

    def normalize(text: str) -> str:
        text = text.lower()
        text = re.sub(r"\b(a|an|the)\b", "", text)
        return text.strip()

    def evaluate_guess(guess: str, target: str) -> bool:
        guess = normalize(guess)
        target = normalize(target)

        if guess == target:
            return True

        if target in SYNONYMS and guess in SYNONYMS[target]:
            return True

        return False

    def image_to_base64(image: Image.Image) -> str:
        buffer = BytesIO()
        image.save(buffer, format="PNG")
        return base64.b64encode(buffer.getvalue()).decode()

    def generate_words():
        model = genai.GenerativeModel(TEXT_MODEL)

        prompt = """
        Generate 10 very simple, single-word noun objects
        that are easy to draw in under 20 seconds.

        Rules:
        - Only common physical objects
        - No abstract concepts
        - No multi-word phrases
        - Output ONLY valid JSON

        JSON format:
        {"words": ["cat", "house", "tree"]}
        """

        response = model.generate_content(
            prompt,
            generation_config={"temperature": 0.6}
        )

        data = safe_json_parse(response.text)
        return data["words"] if data else ["cat", "house", "tree", "car", "cup"]

    def gemini_guess(image_base64: str) -> str:
        try:
            model = genai.GenerativeModel(VISION_MODEL)

            prompt = """
            You are playing a QuickDraw-style guessing game.

            Rules:
            - Look carefully at the drawing.
            - Return exactly ONE lowercase word.
            - No punctuation.
            - No explanations.
            - If unsure, return "uncertain".
            - Output ONLY valid JSON.

            JSON format:
            {"guess":"word"}
            """

            image_part = {
                "mime_type": "image/png",
                "data": base64.b64decode(image_base64)
            }

            response = model.generate_content(
                [prompt, image_part],
                generation_config={"temperature": 0.2}
            )
```

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    data = safe_json_parse(response.text)
    return data["guess"] if data else "uncertain"

except Exception:
    return "uncertain"
```

How to Run the Project

1. Install dependencies:
pip install streamlit streamlit-drawable-canvas pillow google-generativeai

2. Set Gemini API key:
export GEMINI_API_KEY="your_api_key_here"

3. Run the app:
streamlit run app.py

This project demonstrates: • Multimodal GenAI • Prompt Engineering • Vision-based inference • Deterministic AI evaluation • Human-AI interaction loop • Production-grade Streamlit architecture